The Dynamics of Human Skill Acquisition

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Supervisors

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Abstract

Researchers often describe improvement through practice using the Power Law of Practice. However, recent studies suggest that this law may emerge from averaging across many individuals, while single learners may follow different dynamics.

In this project, I analyze individual learning trajectories using longitudinal performance data from competitive Rubik's Cube players. I apply non-linear regression, time-series analysis, and statistical model selection to characterize both long-term learning trends and short-term performance fluctuations. I develop a scalable and reproducible analysis pipeline that allows me to extend the analysis to multiple players and events.

Motivation and Research Questions

Understanding how individuals acquire complex skills plays a central role in cognitive science and complexity science. Aggregate analyses often hide individual variability. Therefore, I focus on competitive speedcubing, which provides high-resolution longitudinal data for single players over many years.

The project addresses the following research questions:

- **The Trend:** Does individual performance improvement follow a Power Law or an Exponential decay when I analyze the data at the single-player level?
- **The Noise:** Do short-term performance fluctuations exhibit long-range correlations, which would indicate memory-dependent learning dynamics?
- **The Distributions:** How does the distribution of solving times evolve as players approach their performance limits?

Methodology and Process

I structure the project according to the CRISP-DM framework and emphasize transparency and reproducibility throughout the analysis.

Data. I use publicly available competition results from the World Cube Association (WCA). This data contains timestamps of competitions, event types (e.g. 3×3×3 cube), individual solve times, averages, and rankings. I initially focus on a limited number of players and one main event and later, I extend the pipeline to additional players to demonstrate scalability.

Data sources:
`https://www.worldcubeassociation.org/export/results`
`https://www.worldcubeassociation.org/persons/2012PARK03?event=333`

Preprocessing.

- I clean the data and remove DNFs (Did Not Finish).
- I order results chronologically for each player.
- I identify competition sessions and construct consistent time indices.

Analysis.

- I fit Power Law and Exponential learning models using non-linear least squares.
- I compare competing models using AIC and BIC to determine which model explains the data better.
- I compute residuals and apply Detrended Fluctuation Analysis (DFA) to estimate the Hurst exponent.
- I analyze how the distribution of solving times changes over different learning phases by fitting Gaussian and heavy-tailed models.

Implementation. I implement the full workflow in reproducible Python notebook(s). The code takes raw competition data as input and outputs cleaned time series, model fits, diagnostics, and visualizations. This structure allows me to apply the same methodology to any other player or event with minimal adjustments.

Expected Results and Evaluation

I expect to obtain:

- A quantitative comparison between Power Law and Exponential learning models at the individual level.
- Evidence for or against long-range correlations in performance fluctuations.
- Clear visualizations of learning trajectories, residual structure, and distributional changes.

To evaluate the models, I use:

- Information criteria (AIC, BIC),
- Residual diagnostics,
- Stability of parameter estimates across different players.

Domain-Specific Lecture

- **Selected Lecture:** 330.214 Project and Enterprise Financing
- **Status:** I complete the lecture in parallel with the project.